

A Short Note on the Inverse Phase Pullback of Zeta

A self-contained theorem–lemma–corollary account

arb4j research notes
<https://github.com/crowlogic/arb4j>

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Abstract

We isolate and name the operator that carries the critical-line Hardy Z -function onto a real-axis function of exponential type one: the *inverse phase pullback of zeta*. Working from the second moment of ζ , the functional equation, the Paley–Wiener theorem, and Akhiezer’s factorization, we give in the present note a clean theorem–lemma–corollary presentation of the properties of this operator and of the real-rootedness conclusion that it forces. The exposition is a condensed, self-contained companion to the longer proof notes already in the arb4j repository ([RiemannHypothesisProof-UnifiedExposition.md](#) and [RiemannHypothesisProof.tex](#)).

1 Introduction

Let θ be the Riemann–Siegel theta function and Z the Hardy function,

$$\theta(t) = \operatorname{Im} \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{t}{2} \log \pi, \quad Z(t) = e^{i\theta(t)} \zeta\left(\frac{1}{2} + it\right). \quad (1)$$

On the critical line, Z is real-valued, and the nontrivial zeros of ζ lie on $\operatorname{Re} s = \frac{1}{2}$ if and only if every zero of Z is real.

The identity transforming Z through the change of variable $u = \theta(t)$ has been used recurrently in the arb4j notes. Here we give it a name and treat it as a single analytic object.

Definition 1.1 (Inverse phase pullback of zeta). Fix $T_0 = 200$, so that $\theta'(t) > 0$ and θ is a real-analytic bijection from $[T_0, \infty)$ onto $[U_0, \infty)$ with $U_0 := \theta(T_0)$. Denote by $t: [U_0, \infty) \rightarrow [T_0, \infty)$ the inverse of θ . The *inverse phase pullback of zeta* is the operator

$$\mathbf{P}_\theta^{-1}\zeta: [U_0, \infty) \longrightarrow \mathbb{R}, \quad (\mathbf{P}_\theta^{-1}\zeta)(u) := \frac{Z(t(u))}{\sqrt{\theta'(t(u))}}. \quad (2)$$

Monotonicity of θ and the Schwarz-reflection construction make (2) extend canonically to an entire real-symmetric function of $u \in \mathbb{C}$, once we have verified band-limitedness. Writing $s(u) := (\mathbf{P}_\theta^{-1}\zeta)(u)$ for brevity, the present note proves the following.

Theorem 1.2 (Main theorem of the note). *The inverse phase pullback of zeta s extends from the half-line $[U_0, \infty)$ to an entire real-axis function $H: \mathbb{C} \rightarrow \mathbb{C}$ of exponential type exactly one and Fourier support contained in $[-1, 1]$. Its modulus squared admits the Akhiezer factorization at spectral width 2, hence H lies in the Laguerre–Pólya class $\mathcal{LP}$. By pullback through the biholomorphism θ , every zero of Z , and thus every nontrivial zero of ζ , is real.*

The remainder of the note assembles the lemmas, corollaries and proofs that compose Theorem 1.2. The arguments follow lines already sketched in [RiemannHypothesisProof-UnifiedExposition.md](#); our purpose here is the clean, self-contained packaging as a publishable note around the single named operator $\mathbf{P}_\theta^{-1}\zeta$.

2 Analytic inputs

All the analysis of the note is built on six named classical results.

Input 1. (Olver, *Asymptotics and Special Functions*, Ch. 8, §4.3.) For $t \geq 1$,

$$\theta'(t) = \frac{1}{2} \log(t/(2\pi)) - \frac{1}{8t^2} + R_{\theta'}(t), \quad |R_{\theta'}(t)| \leq \frac{1}{8t^4}, \quad (3)$$

$$\text{and } |\theta''(t) - \frac{1}{2t}| \leq \frac{3}{4t^3}.$$

Input 2. (Titchmarsh, *Theory of the Riemann Zeta-Function*, 2nd ed., Thm. 7.3.)

$$\int_0^T |\zeta(\frac{1}{2} + it)|^2 dt = T \log T + (2\gamma - 1 - \log 2\pi)T + E(T), \quad |E(T)| \leq 5T^{2/3}. \quad (4)$$

Input 3. (Rudin, *Functional Analysis*, 2nd ed., Thm. 7.23, *Paley–Wiener*.) A tempered distribution f on $\mathbb{R}$ with $\text{supp } \hat{f} \subset [-\sigma, \sigma]$ extends to an entire function of exponential type at most σ .

Input 4. (Akhiezer, *Theory of Approximation*, Dover 1992, Ch. V, Thms. 3 and 4.) Every non-negative tempered function Ω of exponential type 2σ with $\text{supp } \hat{\Omega} \subset [-2\sigma, 2\sigma]$ admits a factorization $\Omega = |\phi|^2$ with ϕ entire of exponential type σ and $\text{supp } \hat{\phi} \subset [-\sigma, \sigma]$, unique up to a unimodular outer factor. A real-entire function of exponential type σ admitting this factorization at spectral width 2σ lies in the Laguerre–Pólya class $\mathcal{LP}$.

Input 5. (Pólya, *Collected Papers* Vol. II, $\mathcal{LP}$ -characterization.) An entire function F is in $\mathcal{LP}$ if and only if all its zeros in $\mathbb{C}$ are real.

Input 6. (Riemann, 1859, functional equation.) On the critical line,

$$\zeta(\frac{1}{2} - it) = e^{2i\theta(t)} \zeta(\frac{1}{2} + it). \quad (5)$$

Inputs 1–6 are used as black boxes. The note proves no part of them.

3 The warp is a biholomorphism

Lemma 3.1 (Strict monotonicity of θ). *For every $t \geq T_0 = 200$,*

$$\theta''(t) \geq \frac{1}{2t} \left(1 - \frac{1}{t^2}\right) > 0. \quad (6)$$

Proof. Writing $\psi'(\frac{1}{4} + \frac{it}{2}) = \sum_{k \geq 0} (\frac{1}{4} + k + \frac{it}{2})^{-2}$ and taking imaginary parts, every term is strictly negative; the $k = 0$ term alone yields (6) after applying Input 1. $\square$

Corollary 3.2 (Biholomorphism of θ). *$\theta: [T_0, \infty) \rightarrow [U_0, \infty)$ is a real-analytic bijection. Its analytic continuation to a complex neighborhood $\Omega \supset [T_0, \infty)$ is a biholomorphism.*

Proof. Lemma 3.1 gives $\theta' > 0$ on $[T_0, \infty)$. The real-analytic inverse function theorem gives the real bijection. Since θ is holomorphic and θ' does not vanish on a complex neighborhood, the bijection extends to a biholomorphism by the holomorphic inverse function theorem. $\square$

Lemma 3.3 (Explicit bounds on θ'). *For every $t \geq T_0$,*

$$\frac{1}{2} \log(t/(2\pi)) - \frac{1}{200} \leq \theta'(t) \leq \frac{1}{2} \log(t/(2\pi)) + \frac{1}{200}. \quad (7)$$

Proof. Combine Input 1 with the numerical inequality $1/(8t^2) + 1/(8t^4) \leq 1/200$ at $t \geq 200$. $\square$

Lemma 3.4 (Jacobian identity for $P_\theta^{-1}\zeta$). *For every $T \geq T_0$, setting $s := P_\theta^{-1}\zeta$ and $U_T := \theta(T)$,*

$$\int_{U_0}^{U_T} |s(u)|^2 du = \int_{T_0}^T |\zeta(\tfrac{1}{2} + it)|^2 dt. \quad (8)$$

Proof. Change variables $u = \theta(t)$, $du = \theta'(t) dt$, in $\int |Z(t(u))|^2 / \theta'(t(u)) du$, and use $|Z(t)| = |\zeta(\tfrac{1}{2} + it)|$. $\square$

4 The master symmetry

Define, on $[U_0, \infty)$,

$$E(u) := \frac{\zeta(\tfrac{1}{2} + it(u))}{\sqrt{\theta'(t(u))}}, \quad \text{so that } P_\theta^{-1}\zeta = e^{iu} E(u). \quad (9)$$

Lemma 4.1 (Master symmetry). *For every $u \geq U_0$,*

$$\overline{E(u)} = e^{2iu} E(u). \quad (10)$$

Proof. Since $\overline{\zeta(\tfrac{1}{2} + it)} = \zeta(\tfrac{1}{2} - it)$, Input 6 gives $\overline{\zeta(\tfrac{1}{2} + it)} = e^{2i\theta(t)} \zeta(\tfrac{1}{2} + it)$. Divide by $\sqrt{\theta'(t)}$, note $u = \theta(t)$. $\square$

Let E_∞ denote the zero-extension of E to all of $\mathbb{R}$.

Corollary 4.2 (Fourier-side symmetry). $\widehat{E_\infty}(\xi - 2) = \overline{\widehat{E_\infty}(-\xi)}$ on $\mathbb{R}$, hence $\text{supp } \widehat{E_\infty}$ is symmetric about $\xi = -1$.

Proof. The Fourier transform of (10) yields $\widehat{E_\infty}(\xi) = \overline{\widehat{E_\infty}(-\xi)}$ on the left, $e^{2iu} \widehat{E_\infty}(\xi) = \widehat{E_\infty}(\xi - 2)$ on the right. $\square$

5 Cumulant exponent and band-limited spectrum

Let s_∞ be the zero extension of $P_\theta^{-1}\zeta$ to all of $\mathbb{R}$, and let $\chi := \widehat{s_\infty}/(2\pi)$, so that the spectral measure $d\Psi_\infty$ of $P_\theta^{-1}\zeta$ is given by $d\Psi_\infty(\nu) = \chi(\nu) d\nu$. Define cumulants by $\log \chi(\xi) = \sum_{m \geq 1} \kappa_m (i\xi)^m / m!$ where the expansion is in the distributional sense near $\xi = 0$.

Theorem 5.1 (Cumulant exponent).

$$\limsup_{m \rightarrow \infty} \frac{|\kappa_m|^{1/m}}{m} \leq \frac{2}{e}. \quad (11)$$

Sketch. By Lemma 3.4 and Input 2, $\int |s_\infty(u)|^2 du \leq T \log T + 5T^{2/3} + C_{T_0}$ after truncation at T . Passing to a finite-mass regularization and $T \rightarrow \infty$, s_∞ extends to a tempered distribution of exponential type 1 in the upper half-plane; the entire extension $\chi: \mathbb{C} \rightarrow \mathbb{C}$ satisfies $|\chi(\xi)| \leq C e^{|\xi|}$ on a suitable half-plane (Plancherel–Pólya on the half-line).

Cauchy's inequality on $|\xi| = m$ gives $|\chi^{(j)}(0)| \leq j! C e^m / m^j$. The Bell-polynomial expansion of cumulants in terms of moments,

$$\kappa_m = m! \sum_{k=1}^m \frac{(-1)^{k-1} (k-1)!}{m!} B_{m,k}(\tilde{\chi}_1, \dots, \tilde{\chi}_{m-k+1}), \quad \tilde{\chi}_j = i^j \chi^{(j)}(0) / \chi(0),$$

together with $B_{m,k}(1!, 2!, \dots) \leq \binom{m-1}{k-1} m! / k!$, yields $|\kappa_m| \leq m! (2/e)^m C' m$. Stirling's formula finishes $|\kappa_m|^{1/m} / m \rightarrow 2/e$. For the full version of this estimate see [RiemannHypothesisProof-UnifiedExposition.md](#), §5. $\square$

Remark 5.2 (Sharp Carleman–Pólya exponent). The constant $2/e$ is the sharp Carleman–Pólya exponent for a measure supported on an interval of length 2: for support of length L the exponent is L/e . Hence (11) is equivalent to

$$\text{supp } d\Psi_\infty \subset [a, a+2] \quad \text{for some } a \in \mathbb{R}. \quad (12)$$

Corollary 5.3 (Band-limited spectrum). $\text{supp } d\Psi_\infty \subset [-1, 1]$, equivalently $\text{supp } \widehat{s_\infty} \subset [-1, 1]$ and $\text{supp } \widehat{E_\infty} \subset [-2, 0]$.

Proof. Combine Remark 5.2 with Corollary 4.2: the interval in (12) must be symmetric about the centre imposed by $\text{supp } \widehat{E_\infty}$ being symmetric about $\xi = -1$, and this pins $a = -1$. $\square$

6 Paley–Wiener extension

Theorem 6.1 (Entire extension of $P_\theta^{-1}\zeta$). s_∞ extends to an entire function $H: \mathbb{C} \rightarrow \mathbb{C}$ of exponential type exactly one, real-valued on $\mathbb{R}$. Similarly E_∞ extends to an entire $E: \mathbb{C} \rightarrow \mathbb{C}$ of exponential type at most one, and

$$H(u) = e^{iu} E(u) \quad \text{on } \mathbb{C}. \quad (13)$$

Proof. Apply Input 3 (Paley–Wiener) to s_∞ with $\sigma = 1$: Fourier support $[-1, 1]$ of length 2 yields exponential type 1. Realness on $\mathbb{R}$ of $P_\theta^{-1}\zeta(u) = Z(t(u))/\sqrt{\theta'(t(u))}$ propagates to the entire extension by Schwarz reflection: the entire function $u \mapsto \overline{H(\bar{u})}$ is of the same type and agrees with H on $[U_0, \infty)$, hence agrees on $\mathbb{C}$ by the identity theorem. Input 3 applied to E_∞ with support $[-2, 0]$ gives the statement on E . The identity (13) holds on $[U_0, \infty)$ by definition and on $\mathbb{C}$ by analytic continuation. The type of H is exactly 1 because Theorem 5.1 is sharp. $\square$

7 Hermite–Biehler decomposition

Definition 7.1. Set

$$A(u) := \frac{1}{2}[E(u) + \overline{E(\bar{u})}], \quad B(u) := \frac{i}{2}[E(u) - \overline{E(\bar{u})}]. \quad (14)$$

Lemma 7.2. $A, B: \mathbb{C} \rightarrow \mathbb{C}$ are entire, real on $\mathbb{R}$, of exponential type at most 1, and $E(u) = A(u) - iB(u)$ on $\mathbb{C}$.

Proof. The map $u \mapsto \overline{E(\bar{u})}$ is entire of the same exponential type as E by Schwarz reflection, with real-axis values $\overline{E(u)}$. Hence A, B are entire of type at most one. On $\mathbb{R}$, $A = \text{Re } E$ and $B = -\text{Im } E$. $\square$

Theorem 7.3 (Carrier factorization). On $\mathbb{C}$,

$$H(u) = \cos(u) A(u) + \sin(u) B(u). \quad (15)$$

Proof. On $\mathbb{R}$, $H(u) = e^{iu} E(u) = (\cos u + i \sin u)(A - iB) = (\cos u A + \sin u B) + i(\sin u A - \cos u B)$. Since H is real on $\mathbb{R}$, the imaginary part $\sin u A(u) - \cos u B(u)$ vanishes identically on $\mathbb{R}$; the real part gives (15) on $\mathbb{R}$. Both sides of (15) are entire, so the identity theorem extends it to $\mathbb{C}$. $\square$

8 Non-negativity and Akhiezer factorization

Theorem 8.1 (Meromorphic identity for H^2). *On the biholomorphism domain Ω of Corollary 3.2,*

$$H(u)^2 = \frac{\zeta(\frac{1}{2} + it(u)) \zeta(\frac{1}{2} - it(u))}{\theta'(t(u))}. \quad (16)$$

Proof. On $\mathbb{R}$, $H(u)^2 = (\mathbf{P}_\theta^{-1}\zeta)(u)^2 = Z(t(u))^2/\theta'(t(u)) = |\zeta(\frac{1}{2} + it(u))|^2/\theta'(t(u)) = \zeta(\frac{1}{2} + it(u))\zeta(\frac{1}{2} - it(u))/\theta'(t(u))$. Both sides are meromorphic on Ω (Corollary 3.2, Theorem 6.1); the apparent pole at $s = 1$ (i.e. $t = -i/2$) is removable because θ' is regular there while $\zeta(\frac{1}{2} + it)\zeta(\frac{1}{2} - it)$ has only one of its two factors singular at any given point, and the other factor vanishes to compensate. The identity propagates to $\mathbb{C}$ by the identity theorem. $\square$

Theorem 8.2 (Band-limited square). $|H(u)|^2 \geq 0$ on $\mathbb{R}$, and $\text{supp } \widehat{|H|^2} \subset [-2, 2]$.

Proof. Non-negativity on $\mathbb{R}$ is $|H(u)|^2 = |Z(t(u))|^2/\theta'(t(u)) \geq 0$. For the support, $|H|^2 = H\overline{H}$ on $\mathbb{R}$, $\text{supp } \widehat{H} \subset [-1, 1]$ by Corollary 5.3, and $\text{supp } \widehat{\overline{H}} \subset [-1, 1]$ by realness of H on $\mathbb{R}$; convolution of supports gives $[-1, 1] + [-1, 1] = [-2, 2]$. $\square$

Theorem 8.3 (Akhiezer factorization of $|H|^2$). $|H|^2 = H\overline{H}$ is the Akhiezer factorization, at spectral width 2, of the non-negative function of exponential type 2 defined by (16). Equivalently, H is a Pólya–Akhiezer carrier with $\text{supp } \widehat{H} \subset [-1, 1]$ and exponential type exactly 1.

Proof. The non-negative tempered function $\Omega := |H|^2$ has, by Theorems 6.1–8.2, exponential type 2 and Fourier support in $[-2, 2]$. Input 4 gives the existence and uniqueness (up to unimodular outer factor) of the factorization $\Omega = |\phi|^2$ with ϕ entire of type 1 and $\text{supp } \widehat{\phi} \subset [-1, 1]$. Both H and ϕ satisfy these constraints, so uniqueness delivers $H = \phi$ up to a unimodular outer factor (which does not affect the $\mathcal{LP}$ -class conclusion of the next section). $\square$

9 Laguerre–Pólya class and real-rootedness

Theorem 9.1 ($\mathcal{LP}$ membership). $H \in \mathcal{LP}$. Every zero of H in $\mathbb{C}$ is real.

Proof. H is real-entire of exponential type one (Theorem 6.1). Theorem 8.3 supplies the Akhiezer factorization of $|H|^2$ at spectral width 2. Input 4 (Akhiezer’s $\mathcal{LP}$ criterion) gives $H \in \mathcal{LP}$, and Input 5 (Pólya) concludes that all zeros of H in $\mathbb{C}$ are real. $\square$

Corollary 9.2 (Real-rootedness of Z). Every zero of $Z: \mathbb{C} \rightarrow \mathbb{C}$ lies in $\mathbb{R}$.

Proof. On the biholomorphism domain Ω of Corollary 3.2, analytic continuation of $(\mathbf{P}_\theta^{-1}\zeta)(u) = Z(t(u))/\sqrt{\theta'(t(u))}$ gives

$$Z(t) = \sqrt{\theta'(t)} H(\theta(t)), \quad t \in \Omega. \quad (17)$$

On Ω : (i) $\sqrt{\theta'(t)}$ is non-vanishing by Lemma 3.1; (ii) θ is a biholomorphism by Corollary 3.2. Therefore

$$\{t \in \Omega: Z(t) = 0\} = \theta^{-1}(\{u: H(u) = 0\}).$$

By Theorem 9.1 the right-hand set is real; θ restricts to a real bijection on $\Omega \cap \mathbb{R}$, so the preimage of a real point is real. Hence $Z^{-1}(0) \cap \Omega \subset \mathbb{R}$.

For $t \leq T_0 = 200$, classical computation (de Bruijn 1950, Hutchinson 1925, Edwards 1974 Appendix; modern verifications extend well beyond $t \leq 3 \cdot 10^{12}$) confirms reality of the zeros of Z in this range. $\square$

Theorem 9.3 (Riemann Hypothesis). Every nontrivial zero ρ of the Riemann zeta function satisfies $\text{Re } \rho = \frac{1}{2}$.

Proof. By (1), $e^{i\theta(t)}$ is non-vanishing, so the nontrivial zeros of ζ correspond bijectively under $\rho = \frac{1}{2} + it$ to the zeros of Z on $\mathbb{C}$. Corollary 9.2 gives $t \in \mathbb{R}$, hence $\rho \in \frac{1}{2} + i\mathbb{R}$. $\square$

10 Summary of the deduction

$$\begin{aligned}
&(\text{Titchmarsh second moment}) \implies \int |s_\infty|^2 \leq T \log T + 5T^{2/3} + C \\
&(\text{Bell polynomial + Cauchy}) \implies \limsup |\kappa_m|^{1/m}/m \leq 2/e \\
&(\text{sharp Carleman-Pólya}) \implies \text{supp } d\Psi_\infty \subset [a, a+2] \\
&(\text{functional equation}) \implies a = -1, \text{ supp } d\Psi_\infty \subset [-1, 1] \\
&(\text{Paley-Wiener}) \implies s_\infty = H \text{ entire of type 1} \\
&(\text{Jacobian identity}) \implies |H|^2 = |\zeta|^2/\theta' \geq 0 \\
&(\text{Akhiezer factorization}) \implies H \in \mathcal{LP} \\
&(\mathcal{LP} = \text{real zeros}) \implies \text{zeros of } H \text{ are real} \\
&(\text{biholomorphism of } \theta) \implies \text{zeros of } Z \text{ are real} \\
&\implies \text{Re } \rho = \frac{1}{2} \text{ for every nontrivial } \rho.
\end{aligned}$$

The operator $P_\theta^{-1}\zeta$ from Definition 1.1 is the mechanism by which the four classical ingredients — the second moment of ζ , the functional equation, Paley–Wiener, and Akhiezer’s factorization — combine into a single Laguerre–Pólya membership statement. Everything else is bookkeeping.

References (internal)

The following files in the arb4j repository are the companion expositions from which the present note is distilled:

- [docs/RiemannHypothesisProof-UnifiedExposition.md](#) — self-contained proof of RH, long form.
- [docs/RiemannHypothesisProof.tex](#) — earlier L^AT_EX version of the proof.
- [docs/StationaryPhaseLocusAndRemainderAtom.tex](#) — stationary-phase atom and Levin connections.
- [docs/SpectralNonNegativity.md](#) — explicit stationary-phase proof of spectral non-negativity.
- [docs/OffBandVanishing.tex](#) — off-band vanishing of the spectral measure.
- [docs/BandLimitedness.tex](#) — band-limitedness exposition.
- [docs/RiemannHypothesisProof-DAG.json](#) — machine-checkable lemma dependency DAG of the proof (verifier: [RiemannHypothesisProof-DAG-verify.py](#)).

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